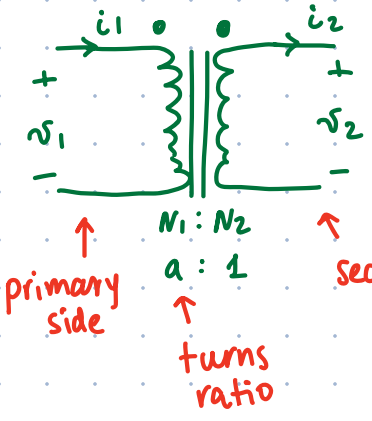
Review 3 (HW 3)

09/21/2026

Ideal transformers, Magnetic equiv. circuits

Ideal transformers:

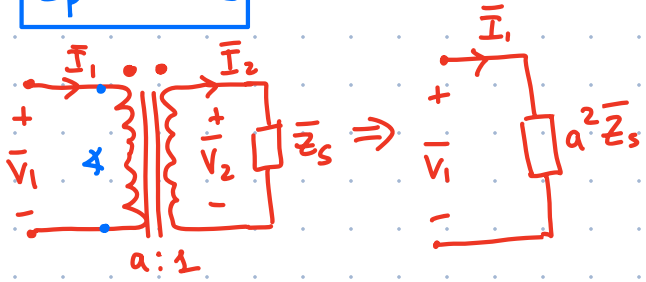


$$\frac{i_1}{i_2} = \frac{N_2}{N_1} = \frac{1}{a} \qquad \frac{v_1}{v_2} = \frac{N_1}{N_2} = a$$

$P_{in} = P_{out}$

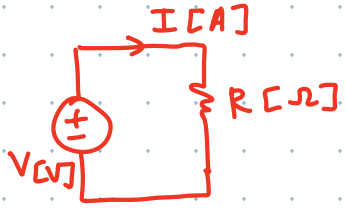
Moving impedance accross a transformer:

$$\bar{Z}_p = a^2 \cdot \bar{Z}_s$$

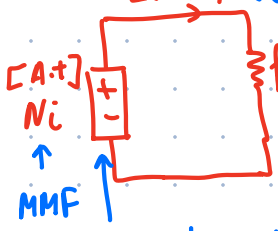


Mag. equiv. circuits:

Electrical



Magnetic



polarity: ϕ flows out of "+" into "−"

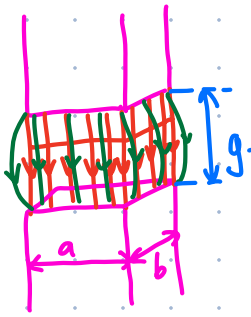
$$\mathcal{R} = \frac{l}{\mu_r \mu_0 A} \qquad (\mu \to \infty,\ \mathcal{R} \to 0)$$

if any of these changes, we need to introduce new reluctance

$\lambda = N\phi = Li$ (flux linkage) [Wb-t]

$v = \frac{d\lambda}{dt} = L\frac{di}{dt}$

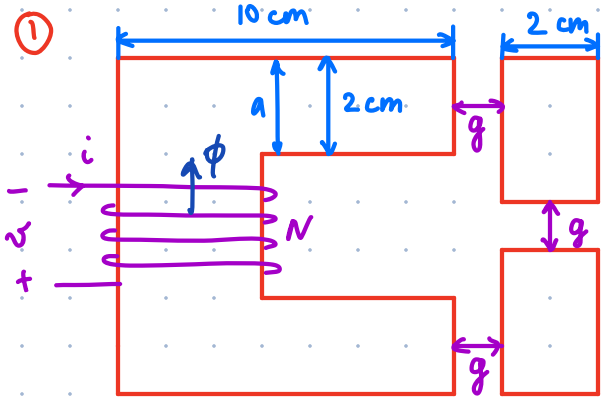
Fringing Effects:



No fringing: $A_g = a \cdot b = A_{core}$

Fringing: $A_g \uparrow$; $A_g = (a+g)(b+g)$

①



$g = 5\,\text{mm} \rightarrow \text{m}$ Fringing effects are included

$N = 250$

$\mu_{core} = \infty \rightarrow k_{core} = 0$

$\mu_0 = 4\pi \cdot 10^{-7}\ H/m$

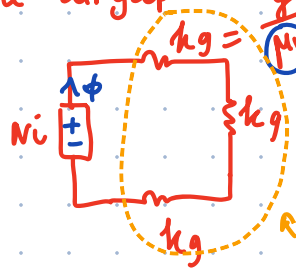
$d\ (\text{depth}) = 3\,\text{cm} \rightarrow \text{m}$

a) Area of the airgap (include fringing)?

$$A = (a+g)(d+g) = (0.02 + 0.005)(0.03 + 0.005) = 8.75 \cdot 10^{-4}\ m^2$$

b) Draw the magnetic equiv. circuit and find the reluctance of one airgap.

$\mathcal{R}_g = \frac{g}{\mu_0 A_g} = 4.54 \cdot 10^6\ \frac{A \cdot t}{wb}$



↑ could combine them into one series reluctance

c) Find inductance of the structure, L.

$$\lambda = L \cdot i = N\Phi = \frac{N^2}{3\mathcal{R}_g} \cdot i \downarrow \quad L = 0.00458\ H$$

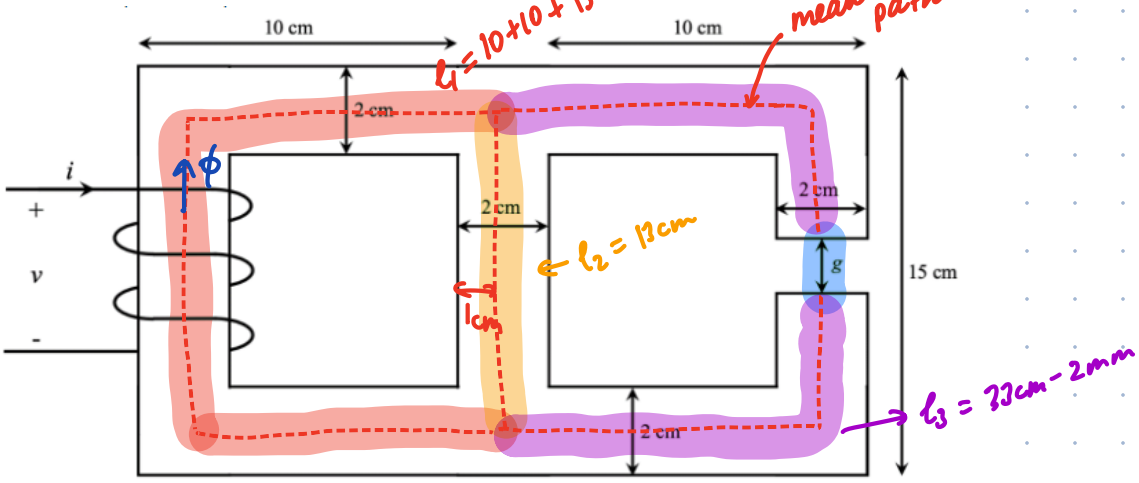
$\Phi = \frac{Ni}{\mathcal{R}_{tot}} = \frac{Ni}{3\mathcal{R}_g} \uparrow \quad \mathcal{R}_g = \frac{g}{\mu_0 A_g} \downarrow$

d) Find inductance of the structure, L_{new}, if fringing is neglected

(L_{new}: smaller)

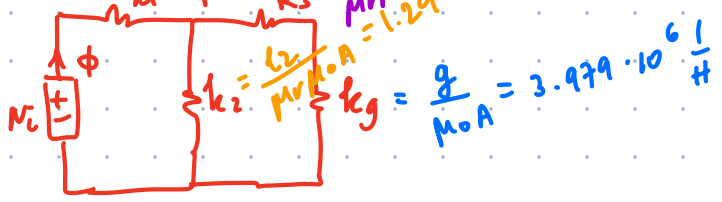
$A_g = a \cdot d = 0.0006\ m^2 \Rightarrow L_{new} = 0.0031\ H$

② Fall 2019, M1, Pr 3



A coil is wrapped 300 times around an iron core with the given dimensions and a depth of 2 cm into the page. The iron has a permeability $\mu = 2000\mu_0$ and contains an air gap of $g = 2$ mm. Fringing effects can be neglected.

a) Draw the magnetic path through the iron core on the figure above and draw the corresponding magnetic equivalent circuit. (15 points).



b) What is the total flux ϕ through the circuit? (7 points)

$$\phi = \frac{Ni}{\mathcal{R}_{tot}} = 6.6 \cdot 10^{-4} \cdot i \text{ Wb}$$

$$\mathcal{R}_{tot} = \mathcal{R}_1 + (\mathcal{R}_2 \,||\, (\mathcal{R}_3 + \mathcal{R}_g))$$

c) What is the inductance of the coil? (3 points)

$$\lambda = L \cdot i = N\phi$$

$$L = 0.198\,H$$